Due: Oct. 26, 2026 @ 11:59 p.m.

General notes to keep in mind:

- *All deliverables for the assignment must be submitted as a **single ZIP file per group** via the Brightspace D2L course shell. Submissions containing multiple ZIP files per group or those with a file that is not in the ZIP format will NOT be graded.*
- *The code submitted must be written purely using the Python programming language and it should execute within the Python 3.12.12 interpreter running on Windows (version 10 or above), Linux or macOS. We recommend running Jupyter inside a virtual environment manager such as pipenv or uv, with the Python version and library versions pinned as specified here. The submitted code should NOT require external python modules other than matplotlib 3.11.2, pandas 3.0.6, seaborn 0.13.2, pywaffle 1.2.0 and their dependencies.*
- *It is your responsibility to ensure that the submitted code executes. If the grader is unable to execute your code and/or your code does NOT adhere to the submission requirements, your code may not be graded. Make sure to run all cells in order before submitting the notebook, as the figures must be reproducible by running all cells top-to-bottom.*
- *The written responses (when required) to the questions in the assignments should be included as a separate cell in the python notebook with the solution.*

EU Citizens on AI and the Gender Bias of Generative AI

Instructions for this assignment

There are many levels of customization that can be accomplished with Matplotlib, Seaborn and PyWaffle. You are free to use all parts of these APIs.

- For this assignment you may import **pandas**, **Matplotlib**, **Seaborn** and **PyWaffle** only.

Data

The datasets `eu-ai.csv` and `gen-ai.csv` are included within the same .zip file.

The `eu-ai.csv` data set has survey results of citizens in EU countries on how AI should be developed. There are 5 rows, representing the survey choices, with responses broken down into individual countries (columns). The survey question was "In your opinion, how important should each of the following actions related to digital technologies be for public authorities?: Shaping the development of Artificial Intelligence and other digital technologies to ensure they respect our rights and values." Responses range from "Very," "Fairly," "Not very," to "Not at all" Important, as well as "Don't know."

The `gen-ai.csv` data set is a gender analysis of 5,100 images of AI-generated people made by the text-to-image model Stable Diffusion v1.5. The model was prompted by asking for high-quality images of people from various occupations. Each image was manually reviewed to label the gender of the person in the image. The data set contains the result of these labelings, providing the number of people per occupation that were classified into men, women, or were ambiguous.

These data sets about AI comes from the following articles and data sources:

- "Humans are Biased. Generative AI is Even Worse. Stable Diffusion's text-to-image model amplifies stereotypes about race and gender – here's why that matters," by Leonardo Nicoletti and Dina Bass, Bloomberg, https://www.bloomberg.com/graphics/2023-generative-ai-bias/
- "EU Data Stories," Nadieh Bremer, https://www.visualcinnamon.com/portfolio/eu-data-stories-2025/
- "A Promising Horizon: How the 2025–2027 Horizon Europe strategic plan will shape Europe's tomorrow," European Data, https://data.europa.eu/en/publications/datastories/promising-horizon-how-2025-2027-horizon-europe-strategic-plan-will-shape

- "Special Eurobarometer SP551: The Digital Decade," European Data, https://data.europa.eu/data/datasets/s3174_101_2_sp551_eng

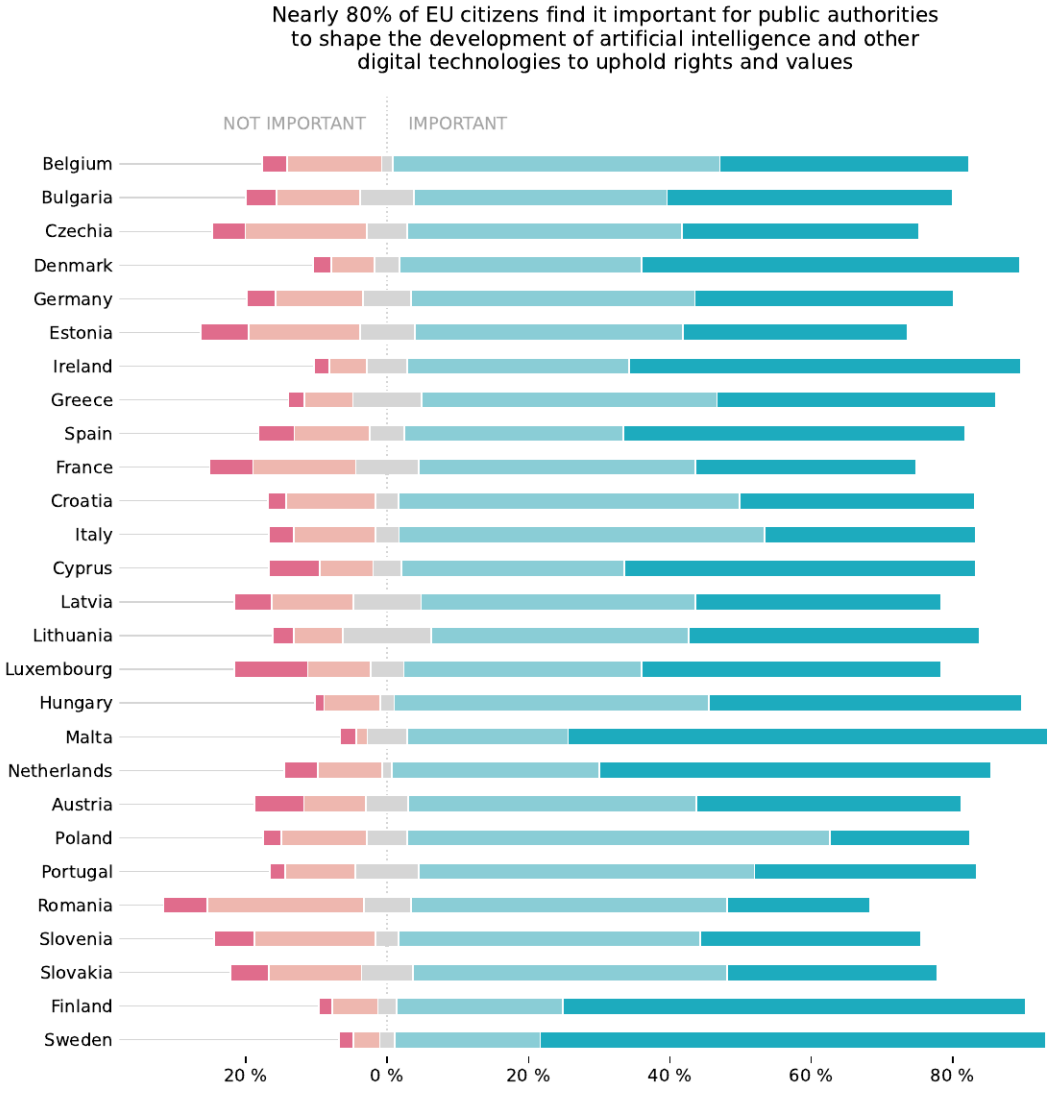
Question 1 - Diverging Bar Plot - How important is it to shape AI development? [40 marks]

A diverging bar plot is a variant that has bars extending in either direction around some baseline value. For example, if the categories have negative and positive values representing the amount of disagreement or agreement. The baseline value is commonly zero, but the bars could be diverging around some other value relevant to the data set, such as a mean value.

The visualization below shows a diverging bar plot of the survey results from the `eu-ai.csv` data set. People are generally in favour of AI upholding their rights and values.

Re-create this diverging bar plot, **with the same content**, using Matplotlib and Seaborn. The bars should be relative to the zero baseline. The "Don't know" category is centred across this zero baseline. Make sure to represent the aesthetic elements, such as the white gaps delineating each category, the horizontal guide lines (which are only on the left-hand side), the title and other annotations, the ordering of countries, and the vertical dotted line for the baseline value.

The colours needed are #02C0D9, #7DD7E3, #D5D5D5, #A5CEEE, #F9AFA4, and #F35984.



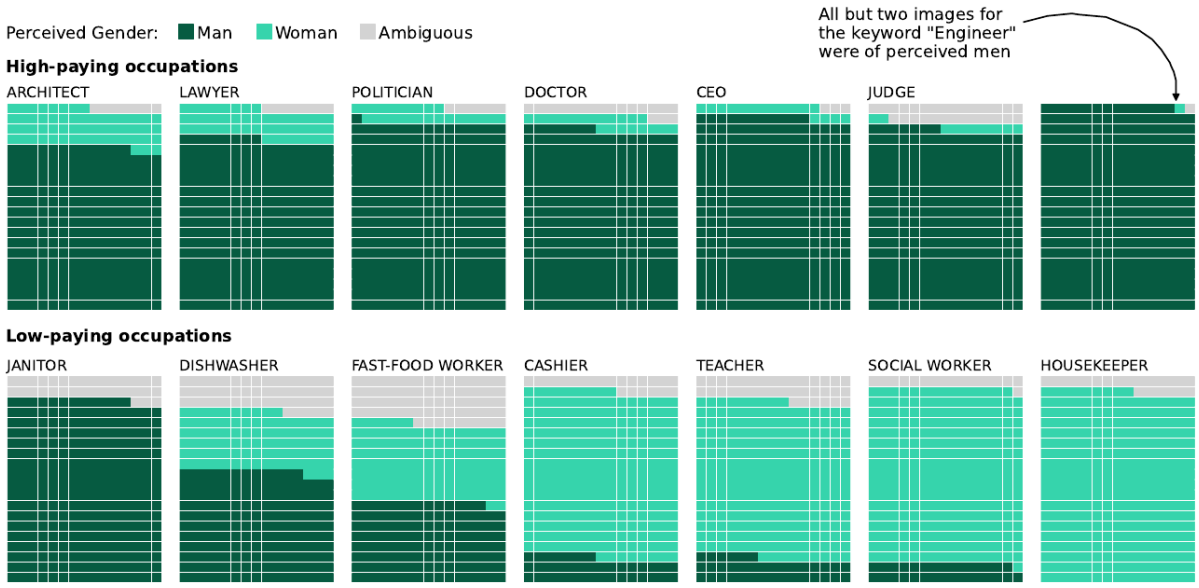
Also, make sure to answer the following question in a separate cell of the notebook: Describe how public opinion on shaping AI development varies across EU countries, and explain, in your vision, why a diverging bar plot communicates this better than a standard (single-direction) stacked bar chart.

Question 2 - Waffle Chart - Gender breakdown of AI-generated people per occupation [60 marks]

The visualization below is a waffle chart showing the breakdown of gender per occupation from the `gen-ai.csv` data set. There is a waffle per occupation, each of which show the gender breakdown of images generated for that occupation.

Recreate the visualization below, **with the same content**, using Matplotlib and PyWaffle. The colours for visualization are #065B41, #36D4AC, and #D3D3D3. The waffles should be sized to be 20×15 (that is, each square represents one data point). Each square of the waffle should be a true square, meaning that they should have the same height and width. The components of each waffle should be ordered as given in the image (men on bottom, ambiguous on top). There should be white lines separating each square of the waffle, with the same spacing between squares in the horizontal and vertical directions.

All text labels should be present, in the correct locations, cases and font weight. The legend should be present above the waffle charts. The text about the engineer waffle should be present, along with the curved line and arrow.



Also, make sure to answer the following question in a separate cell of the notebook: Describe the difference in the gender breakdown of AI-generated people between high-paying and low-paying occupations, and explain why a waffle chart communicates this better than a pie chart per occupation.

Note on grading and overall tips

The grading for Question 1 and Question 2 will be based on the appropriateness of the submitted code and the written responses when requested. Submissions should include an attribution section indicating any sources of material, ideas or contribution of others to the submission. Avoid academic penalties by properly attributing any contribution to your submission by others, including internet sources and classmates. This will also help distinguish what elements of the submission are original. You may not receive full credit if your original elements are insufficient, but you can avoid penalties for plagiarism or copying if you acknowledge your sources.